

DATA*6200 F25 Assignment1

AUTHOR

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1 Introduction

In this project, we analyze the factors that influence income using the **ask_a_manager.xlsx** dataset. In particular, we examine how industry, age, work experience, and education affect income levels. This report is organized into two main sections: **Data Cleaning** and **Visualization and Analysis**.

2 Preparation

Before working with the data, we first install and load the necessary R packages for this project:

```
# install.packages("corrplot")
# install.packages("ggpp")
# install.packages("gridExtra")
# install.packages("readxl")
# install.packages("tidyverse")
library(corrplot)
library(ggpp)
library(glue)
library(grid)
library(gridExtra)
library(readxl)
library(scales)
library(tidyverse)
```

3 Data Cleaning

3.1 Preprocessing

Before cleaning specific target columns, we begin by reorganizing the dataframe and handling any invalid or inconsistent records that could affect the overall analysis.

First, we load the dataset into R as a dataframe:

```
salaries_df <- read_excel("ask_a_manager.xlsx")
```

Next, we perform a quick inspection to check the size of the dataframe and review its column names:

► Code

```
the number of columns: 18
the number of rows: 28080
column names:
"Timestamp"
```

"How old are you?"

"What industry do you work in?"

"Job title"

"If your job title needs additional context, please clarify here:"

"What is your annual salary? (You'll indicate the currency in a later question. If you are part-time or hourly, please enter an annualized equivalent -- what you would earn if you worked the job 40 hours a week, 52 weeks a year.)"

"How much additional monetary compensation do you get, if any (for example, bonuses or overtime in an average year)? Please only include monetary compensation here, not the value of benefits."

"Please indicate the currency"

"If 'Other,' please indicate the currency here:"

"If your income needs additional context, please provide it here:"

"What country do you work in?"

"If you're in the U.S., what state do you work in?"

"What city do you work in?"

"How many years of professional work experience do you have overall?"

"How many years of professional work experience do you have in your field?"

"What is your highest level of education completed?"

"What is your gender?"

"What is your race? (Choose all that apply.)"

It appears that the original survey questions were used as column names, which makes data processing cumbersome. To simplify the workflow, we rename these columns and add two new columns, **id** and **valid**, for convenience during analysis:

► Code

[1] "id"	"valid"	"timestamp"
[4] "age"	"industry"	"job_title"
[7] "job_title_comments"	"annual_salary"	"extra_compensation"
[10] "currency1"	"currency2"	"income_context"
[13] "country"	"us_state"	"city"
[16] "years"	"years_in_field"	"education"
[19] "gender"	"race"	

At this stage, we also examine the data types of each column to ensure they are appropriate:

► Code

```
id: integer
valid: logical
timestamp: POSIXct
timestamp: POSIXt
age: character
industry: character
job_title: character
job_title_comments: character
annual_salary: numeric
extra_compensation: character
currency1: character
currency2: character
income_context: character
country: character
```

```

us_state: character
city: character
years: character
years_in_field: character
education: character
gender: character
race: character

```

Before cleaning individual columns, we exclude unwanted or invalid records. Since the data was collected from random respondents, some entries may contain unrealistic or intentionally misleading information. To identify such cases, we inspect columns with many missing values (e.g., **currency2**), as rows with numerous null entries often contain unreliable data:

► Code

id	valid	timestamp	age	
<int>	<lgl>	<dtm>	<chr>	
422	TRUE	2021-04-27 11:09:44.019000	45-54	
435	TRUE	2021-04-27 11:09:53.862999	25-34	
604	TRUE	2021-04-27 11:12:23.954999	45-54	
753	TRUE	2021-04-27 11:14:29.470000	45-54	
767	TRUE	2021-04-27 11:14:39.157999	25-34	
777	TRUE	2021-04-27 11:14:45.141000	45-54	
1002	TRUE	2021-04-27 11:18:26.555000	35-44	
1218	TRUE	2021-04-27 11:21:56.776000	25-34	
1312	TRUE	2021-04-27 11:23:38.944999	25-34	
1841	TRUE	2021-04-27 11:33:16.266999	35-44	

1-10 of 217 rows | 1-4 of 20 columns Previous **1** [2](#) [3](#) [4](#) [5](#) [6](#) ... [22](#) [Next](#)

The records with id = **28008**, **28010**, **28016**, **28022**, and **28051** were clearly invalid and irrelevant to the analysis. These records are therefore marked as invalid:

► Code

id	valid	timestamp	age	
<int>	<lgl>	<dtm>	<chr>	
28008	FALSE	2024-02-06 21:15:27.638999	18-24	
28010	FALSE	2024-02-10 15:04:31.440000	25-34	
28016	FALSE	2024-02-22 07:27:47.193000	18-24	
28022	FALSE	2024-03-20 13:20:31.025000	18-24	
28050	FALSE	2024-06-28 13:59:59.957000	25-34	

5 rows | 1-4 of 20 columns

With these steps completed, the preprocessing phase is done. In the next sections, we will clean and prepare the columns directly related to the core analysis tasks.

3.2 Currency

The most crucial component of this dataset is the income information. However, we cannot directly compare compensation values in their raw form because they are expressed in different currencies. Therefore, the first step is to organize and standardize the currency information.

Currently, currency data is stored across two columns: **currency1** and **currency2**. While **currency1** usually contains valid information, **currency2** is used when "Other" is selected in **currency1**. To consolidate this information, we create a new column, **currency**, which combines data from both fields:

► Code

currency	
<chr>	
AMERICAN DOLLARS	
ARGENTINE PESO	
ARGENTINIAN PESO (ARS)	
ARS	
AUD	
AUD AUSTRALIAN	
AUD/NZD	
AUSTRALIAN DOLLARS	
BDT	
BR\$	
1-10 of 76 rows	
Previous 1 2 3 4 5 6 ... 8 Next	

Since **currency2** is a free-text field, its values appear in various formats. Because **currency1** follows the **ISO 4217 three-letter currency code**, we convert all entries in the new currency column to this standard format:

► Code

currency	
<chr>	
ARS	
AUD/NZD	
BDT	
BRL	
CAD	
CHF	
CNY	
COP	
CZK	
DKK	

1-10 of 37 rows

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Next, we handle missing values in the **currency** column:

► Code

id	valid	timestamp	age	
<int>	<lgl>	<dtm>	<chr>	
9345	TRUE	2021-04-27 17:44:21.776	45-54	
18844	TRUE	2021-04-29 03:50:24.400	18-24	
18883	TRUE	2021-04-29 04:30:05.127	25-34	
18905	TRUE	2021-04-29 04:59:14.128	45-54	
27960	TRUE	2023-04-29 15:24:07.341	18-24	

5 rows | 1-4 of 21 columns

Although the exact currencies for these missing records are unknown, we can infer them from the **respondents' work countries**. While this approach is not perfectly accurate, the inferred currency values seem reasonable based on the reported income amounts. To retain as many records as possible for analysis, we fill in missing currencies according to the country of employment:

► Code

The number of nulls in the currency column: 0

At this stage, we notice a few special cases where **currency1** is not "Other," yet **currency2** still contains relevant information:

► Code

id	valid	timestamp	age	
<int>	<lgl>	<dtm>	<chr>	
422	TRUE	2021-04-27 11:09:44.019000	45-54	
753	TRUE	2021-04-27 11:14:29.470000	45-54	
767	TRUE	2021-04-27 11:14:39.157999	25-34	
777	TRUE	2021-04-27 11:14:45.141000	45-54	
1002	TRUE	2021-04-27 11:18:26.555000	35-44	
1218	TRUE	2021-04-27 11:21:56.776000	25-34	
1916	TRUE	2021-04-27 11:34:43.130000	35-44	
2049	TRUE	2021-04-27 11:36:58.039000	25-34	
2874	TRUE	2021-04-27 11:53:51.713000	35-44	
3125	TRUE	2021-04-27 11:59:25.947000	25-34	

1-10 of 56 rows | 1-4 of 21 columns

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For the records with **id** = **1916, 3606, 11658, 14074, 25614, 27975**, and **28031**, the currency

information in **currency2** should be used instead of **currency1**. We update those entries accordingly:

► Code

currency
<chr>
ARS
AUD/NZD
BDT
BRL
CAD
CHF
CNY
COP
CZK
DKK

1-10 of 38 rows

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As the final step in cleaning the currency data, we address records labeled **AUD/NZD**:

► Code

id	valid	timestamp	age
<int>	<lgl>	<dtm>	<chr>
140	TRUE	2021-04-27 11:05:50.644999	35-44
471	TRUE	2021-04-27 11:10:22.720999	35-44
1189	TRUE	2021-04-27 11:21:33.509999	25-34
1363	TRUE	2021-04-27 11:24:31.190999	35-44
2369	TRUE	2021-04-27 11:42:57.756000	35-44
4008	TRUE	2021-04-27 12:22:20.996000	35-44
4221	TRUE	2021-04-27 12:28:09.884999	35-44
4367	TRUE	2021-04-27 12:31:53.480000	35-44
4527	TRUE	2021-04-27 12:37:40.849999	18-24
7201	TRUE	2021-04-27 14:33:19.427000	35-44

1-10 of 511 rows | 1-4 of 21 columns

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The original dataset includes an **AUD/NZD** option under **currency1**, but Australian Dollar (AUD) and New Zealand Dollar (NZD) are distinct currencies with different exchange rates. Therefore, we split this combined category into separate AUD and NZD currencies.

By reviewing the records, we find that most respondents with AUD/NZD entries are based in Australia or New Zealand, so assigning currencies based on workplace location is reasonable. However, two records have AUD/NZD listed as their currency while their workplaces are outside of Australia and New Zealand:

► Code

id	valid	timestamp	age
<int>	<lgl>	<dtm>	<chr>
17427	TRUE	2021-04-28 20:25:20.289000	45-54
25511	TRUE	2021-05-06 18:12:15.734999	25-34

2 rows | 1-4 of 21 columns

Since their true currencies cannot be determined confidently, we exclude those two records. For all other AUD/NZD entries, we assign the currency based on the respondent's workplace:

► Code

currency
<chr>
ARS
AUD
BDT
BRL
CAD
CHF
CNY
COP
CZK
DKK

1-10 of 39 rows

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With these adjustments, the currency information is now ready for analysis. To enable income comparison across all records, we add **exchange rate** data and convert all income values to a common currency, **USD**. For convenience, we also include **currency descriptions** alongside their respective codes:

► Code

currency	currency_desc	currency_rate
<chr>	<chr>	<dbl>
ARS	Argentine Peso	0.00074
AUD	Australian Dollar	1.00000
BDT	Bangladeshi Taka	0.00820
BRL	Brazilian Real	0.18000
CAD	Canadian Dollar	0.71000
CHF	Swiss Franc	1.24000
CNY	Chinese Yuan	0.14000
COP	Colombian Peso	0.00025
CZK	Czech Koruna	0.04800

currency	currency_desc	currency_rate
<chr>	<chr>	<dbl>
DKK	Danish Krone	0.15000

1-10 of 39 rows

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3.3 Income

Now that the currency information has been organized, we can apply the exchange rates to the income data. Before proceeding, however, we will handle the missing values:

► Code

The number of nulls

- annual_salary: 0
- extra_compensation: 7303

There are no null values in the **annual_salary** column, but many in **extra_compensation**. Since the missing values in **extra_compensation** likely indicate that respondents did not receive any additional pay, we will replace the nulls with **zeros**. We will also convert the **extra_compensation** column from a character type to an integer for proper calculation:

► Code

The number of nulls

- annual_salary: 0
- extra_compensation: 0

The data types

- annual_salary: numeric
- extra_compensation: integer

With the missing values resolved, we can now apply the exchange rates to the income columns. We will create two new columns, **annual_salary_usd** and **extra_compensation_usd**, and calculate the total as **annual_income_usd**:

► Code

id	valid	timestamp	age
<int>	<lgl>	<dtm>	<chr>
1	TRUE	2021-04-27 11:02:09.743000	25-34
2	TRUE	2021-04-27 11:02:21.562000	25-34
3	TRUE	2021-04-27 11:02:38.125000	25-34
4	TRUE	2021-04-27 11:02:40.642999	25-34
5	TRUE	2021-04-27 11:02:41.792999	25-34
6	TRUE	2021-04-27 11:02:45.571000	25-34

6 rows | 1-4 of 26 columns

3.4 Industry

We will analyze the relationship between industry and salary as the first task, so let's first examine and organize the **industry** column.

Looking at the raw data:

► Code

industry	
<chr>	
"Government Relations" (Lobbying)	
Academia	
Academia - STEM	
Academia / Research	
Academia--cell and molecular biology	
Academic Medicine	
Academic Press Production	
Academic Publishing	
Academic Science	
Academic Scientific Research	
1-10 of 1,132 rows	
Previous 1 2 3 4 5 6 ... 114 Next	

There are over 1,000 distinct industry entries, which is too many for meaningful comparison. Therefore, we need to consolidate and group them.

First, let's fix inconsistencies in the text formatting of industry names. The variations include:

- Presence or absence of double quotation marks
- Use or omission of spaces
- Use of hyphens
- "and" vs. "&"

To improve consistency, we will

- Remove double quotation marks
- Remove hyphens
- Replace "and" with "&"
- Eliminate multiple consecutive spaces

We will apply these modifications to create a new column **industry_mod**:

► Code

industry_mod	
<chr>	
academia	

industry_mod	
<chr>	
academia cell & molecular biology	
academia stem	
academia / research	
academic / nonprofit research	
academic medicine	
academic press production	
academic publishing	
academic research	
academic research (psychology)	
1-10 of 996 rows	
Previous 1 2 3 4 5 6 ... 100 Next	

After cleaning, the number of unique industry entries decreases to **996**. Next, we will manually categorize them into broader groups to simplify analysis.

Since categorization is inherently subjective, others may classify these differently. However, for this project, I have reviewed all entries and organized them into **59 industry categories** (see the code for details). Additionally, I created special categories for cases better excluded from the main analysis:

- multiple jobs
- self employed / entrepreneur
- unemployment

Industries that did not fit neatly into any category were grouped as **"Other"**, and records without industry information were labeled **"Unclear"**. These new categories are stored in the **industry_group** column:

► Code

industry_group	
<chr>	
#accounting	
#administration	
#agriculture	
#animal health & welfare	
#architectural, engineering, & construction	
#arms	
#art	
#association management	
#automotive	
#aviation	
1-10 of 65 rows	
Previous 1 2 3 4 5 6 7 Next	

The **industry_group** column still contains some null values, and since there are no clear indicators

for classification, we will exclude those records from the analysis. (We could assign them to “**Unclear**,” but keeping them distinct helps preserve analytical transparency.)

► Code

id	valid	timestamp	age	industry	industry_mo
<int>	<lgl>	<dtm>	<chr>	<chr>	<chr>
361	TRUE	2021-04-27 11:08:41.197000	45-54	NA	NA
646	TRUE	2021-04-27 11:12:58.989000	35-44	NA	NA
1605	TRUE	2021-04-27 11:28:19.789999	25-34	NA	NA
2056	TRUE	2021-04-27 11:37:01.190000	35-44	NA	NA
2064	TRUE	2021-04-27 11:37:11.960000	35-44	NA	NA
2105	TRUE	2021-04-27 11:38:00.397000	25-34	NA	NA
2109	TRUE	2021-04-27 11:38:03.743999	25-34	NA	NA
2435	TRUE	2021-04-27 11:43:58.180999	25-34	NA	NA
4078	TRUE	2021-04-27 12:24:20.198999	25-34	NA	NA
4679	TRUE	2021-04-27 12:43:23.078999	18-24	NA	NA

1-10 of 74 rows | 1-6 of 28 columns Previous **1** [2](#) [3](#) [4](#) [5](#) [6](#) ... [8](#) [Next](#)

3.5 Age

In this project, we also aim to explore how salary relates to **age**, **work experience (years)**, **experience in the field (years_in_field)**, and **education**. Let's start by cleaning the **age** data.

The **age** column is already well organized and contains no null records. For sorting purposes, we will create a new column, **age_sort**:

► Code

age	age_sort
<chr>	<chr>
under 18	00-17
18-24	18-24
25-34	25-34
35-44	35-44
45-54	45-54
55-64	55-64
65 or over	65-

7 rows

No additional cleaning is required for this column.

3.6 Professional Work Experience

Both the **years** and **years_in_field** columns contain no null values. However, the category labels show some inconsistencies in formatting:

► Code

years

<chr>

1 year or less

11 - 20 years

2 - 4 years

21 - 30 years

31 - 40 years

41 years or more

5-7 years

8 - 10 years

8 rows

► Code

years_in_field

<chr>

1 year or less

11 - 20 years

2 - 4 years

21 - 30 years

31 - 40 years

41 years or more

5-7 years

8 - 10 years

8 rows

We will standardize the format of these columns directly and create new columns for sorting purposes:

► Code

years

<chr>

1 year or less

2-4 years

5-7 years

8-10 years

11-20 years

21-30 years

31-40 years

41 years or more

years_sort

<chr>

00-01

02-04

05-07

08-10

11-20

21-30

31-40

41-

8 rows

► Code

years_in_field	years_in_field_sort
<chr>	<chr>
1 year or less	00-01
2-4 years	02-04
5-7 years	05-07
8-10 years	08-10
11-20 years	11-20
21-30 years	21-30
31-40 years	31-40
41 years or more	41-
8 rows	

3.7 Education

The categories in the **education** column are well organized. However, some records contain null values. Since there is no reliable way to infer the missing education levels, we will exclude these records from the analysis:

► Code

education
<chr>
College degree
High School
Master's degree
PhD
Professional degree (MD, JD, etc.)
Some college
NA
7 rows

► Code

The number of nulls: 222

4 Visualization and Analysis

4.1 Industry vs Income

As the first task, we aim to examine which **industry groups** tend to have higher income levels. We begin by calculating the average income for each industry group:

► Code

industry_group	annual_income_usd
<chr>	
#accounting	24569
#fitness	1678
*multiple jobs	1410
#tech	1394
#energy	135
#biology	125
#legal	125
#trade	122
#storage facility	1200
#sales	118

1-10 of 65 rows | 1-2 of 3 columns

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Upon inspection, the **accounting** group shows an abnormally high average income. Let's look into the corresponding records:

► Code

id	valid	annual_salary	annual_income_usd	currency	currency_rate
<int>	<lgl>	<dbl>	<dbl>	<chr>	<dbl>
28056	TRUE	6000070000	4260049700.00	CAD	0.71000
26936	TRUE	450000	1950000.00	USD	1.00000
18280	TRUE	215000	1615000.00	USD	1.00000
18323	TRUE	215000	1615000.00	USD	1.00000
14524	TRUE	850000	850000.00	USD	1.00000
15310	TRUE	800000	800000.00	USD	1.00000
16965	TRUE	300000	750000.00	USD	1.00000
19516	TRUE	250000	750000.00	USD	1.00000
15818	TRUE	250000	691600.00	GBP	1.33000
28031	TRUE	5000000	650005.20	HKD	0.13000

1-10 of 1,811 rows | 1-6 of 7 columns

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The record with **id = 28056** has an unrealistically large salary value, which clearly appears to be a typo. Considering similar records in the same industry, it is reasonable to assume the intended amount might be around **60,000–70,000 CAD**:

► Code

id	valid	annual_salary	annual_income_usd	currency	currency_rate
<int>	<lgl>	<dbl>	<dbl>	<chr>	<dbl>
28056	TRUE	6000070000	4260049700.00	CAD	0.71000
7125	TRUE	65000	49700.00	CAD	0.71000
11147	TRUE	65000	49700.00	CAD	0.71000
13331	TRUE	70000	49700.00	CAD	0.71000
13432	TRUE	70000	49700.00	CAD	0.71000
12708	TRUE	48763	49663.00	USD	1.00000
2707	TRUE	49000	49500.00	USD	1.00000
2790	TRUE	49000	49500.00	USD	1.00000
4127	TRUE	67500	49238.50	CAD	0.71000
22328	TRUE	33000	49210.00	GBP	1.33000

1-10 of 311 rows | 1-6 of 21 columns

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Although we could manually adjust the salary to, for example, **65,000 CAD**, we will instead remove this record to maintain data integrity, as the true value is uncertain and the dataset contains sufficient accounting records:

► Code

industry_group	annual_income_usd
<chr>	
#fitness	1678
*multiple jobs	1410
#tech	1394
#energy	135
#biology	125
#legal	125
#trade	122
#storage facility	1200
#sales	118
#consulting	1125

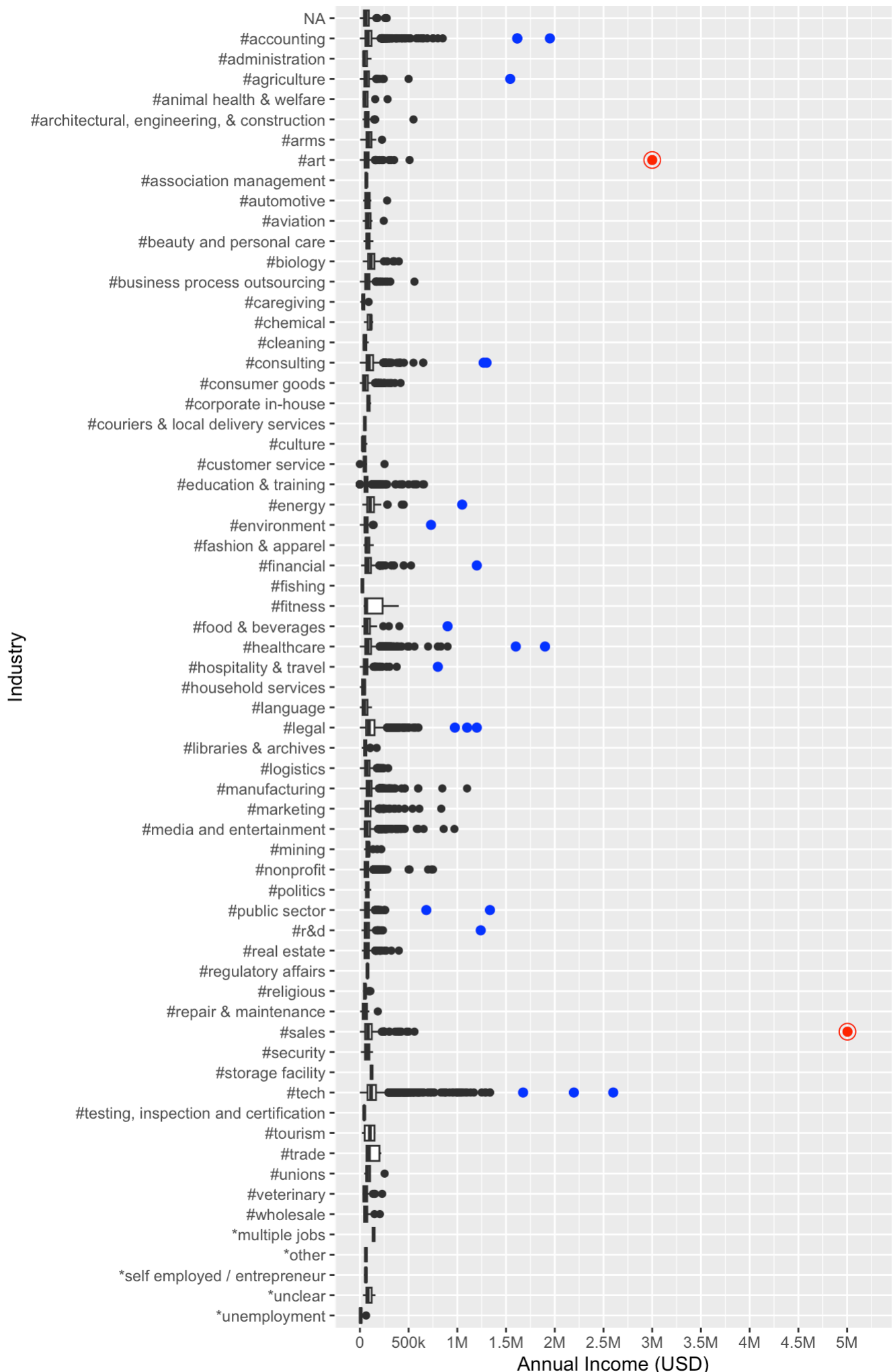
1-10 of 65 rows | 1-2 of 3 columns

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After this adjustment, the average income per industry group appears reasonable. Next, we will visualize the income distributions by industry group:

► Code

Annual Income Distribution by Industry



From the distribution plot, several records stand out with unusually high income values.

- Red points represent extreme outliers that are significantly higher than the rest within the same industry - we will refer to these as **Level 1 outliers**.
- Blue points represent moderately high outliers that are still noticeably higher than other records in their group — we will call these **Level 2 outliers**.

Upon reviewing the Level 1 and Level 2 outliers, most belong to individuals holding top executive positions (e.g., CEOs) or earning substantial amounts from stock options:

► Code

id	age	industry_group	annual_income_usd	currency
<int>	<chr>	<chr>	<dbl>	<chr>
26936	35-44	#accounting	1950000	USD
18280	25-34	#accounting	1615000	USD
18323	25-34	#accounting	1615000	USD
5582	18-24	#agriculture	1542800	GBP
2125	55-64	#art	3000000	USD
26559	45-54	#consulting	1300000	USD
17709	25-34	#consulting	1270000	USD
14073	35-44	#energy	1049370	GBP
26723	25-34	#environment	730000	AUD
19453	35-44	#financial	1200000	USD

1-10 of 24 rows | 1-5 of 20 columns

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A key question arises: **should we exclude only Level 1 outliers, or both Level 1 and Level 2?** By comparing the top six industries ranked by median and mean income under both exclusion conditions, we find that removing Level 2 outliers has little effect on the median, but it does alter the mean rankings. Notably, a single Level 2 record in the **energy** industry raises its mean rank from **6th to 3rd**, a significant distortion.

Since most Level 1 and Level 2 outliers represent unusual financial situations, we will proceed with the **Level 1 & Level 2–excluded** results:

► Code

```
Top 6 industries (level1-excluded, median):
list(industry_group = c("#storage facility", "#tech", "#biology", "#chemical",
"#energy", "#tourism"))
```

```
Top 6 industries (level1&2-excluded, median):
list(industry_group = c("#storage facility", "#tech", "#biology", "#chemical",
"#energy", "#tourism"))
```

```
Top 6 industries (level1-excluded, mean):
list(industry_group = c("#fitness", "#tech", "#energy", "#biology", "#legal",
```

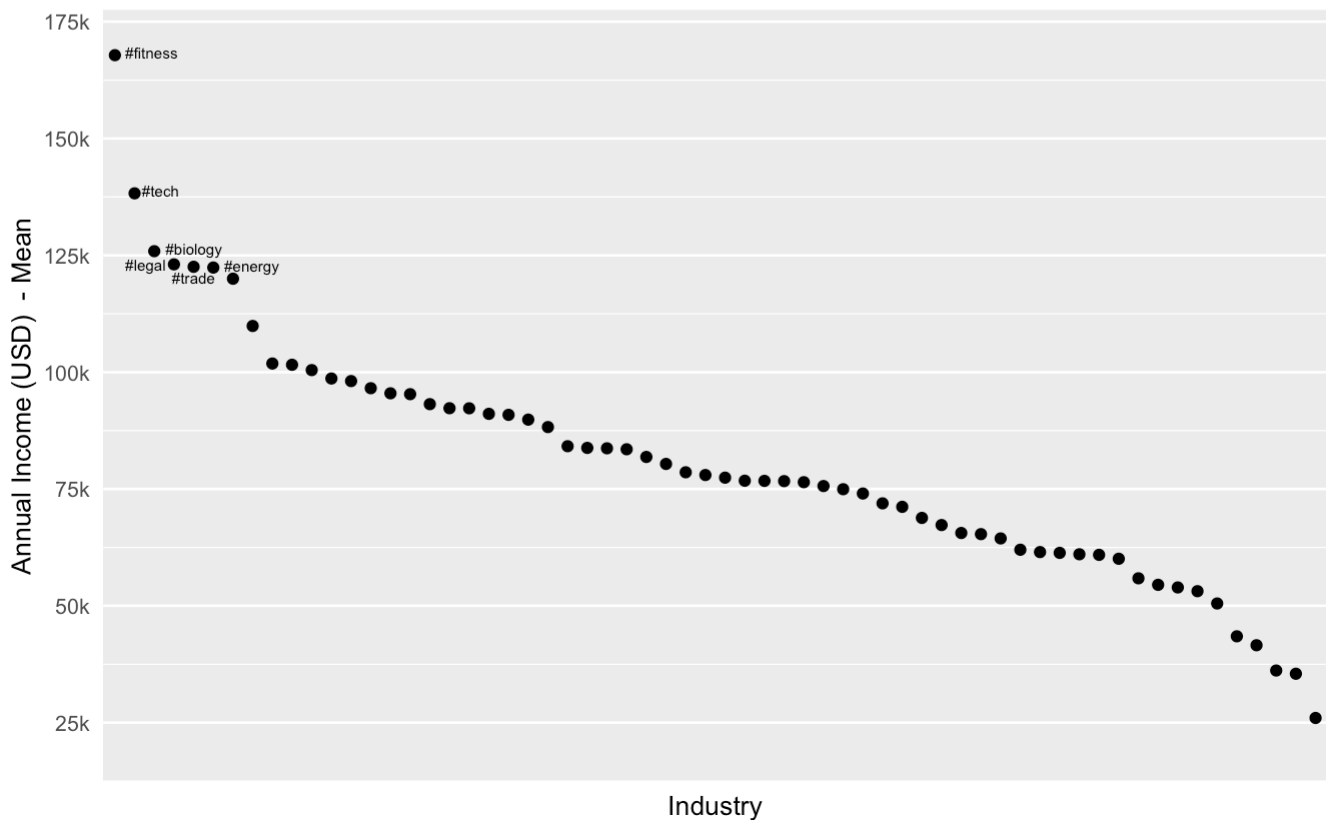
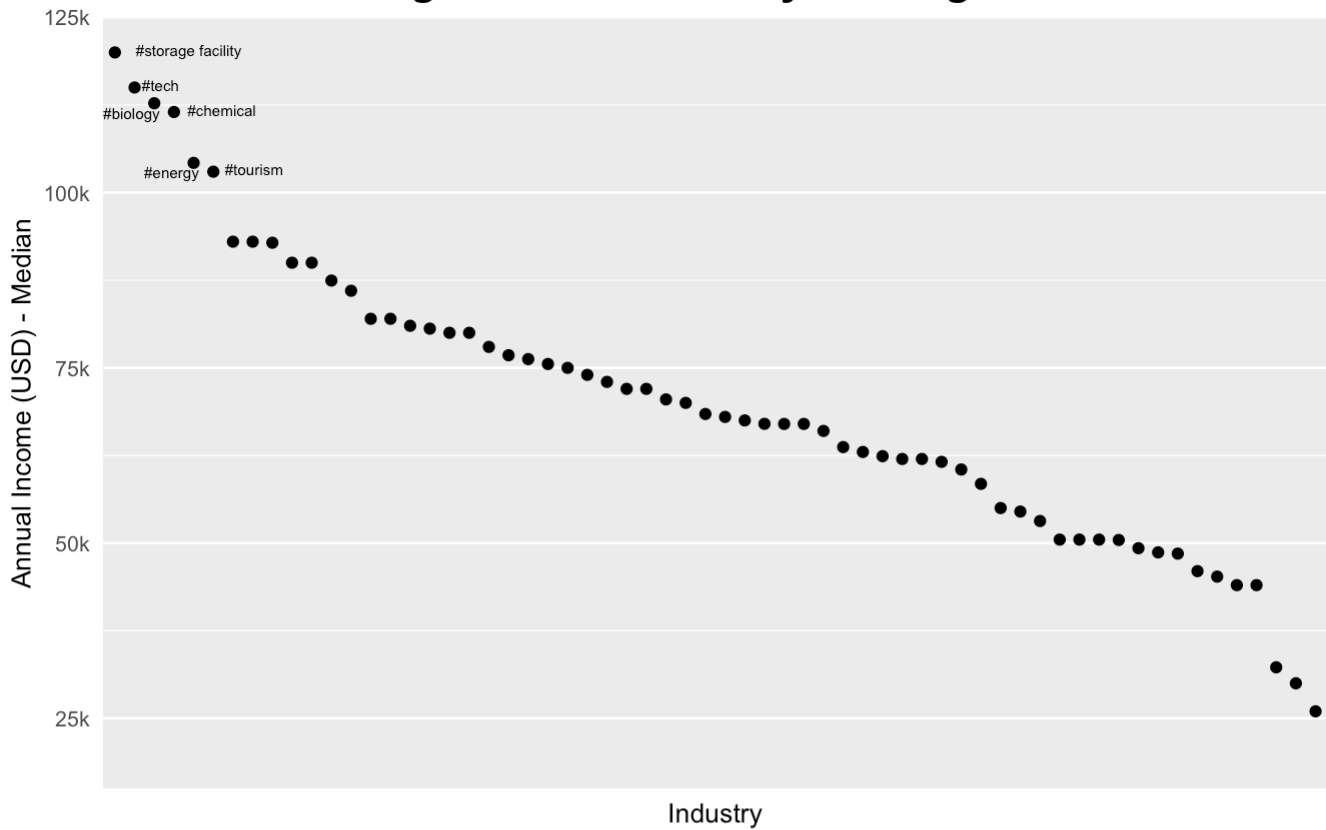
```
"#trade"))
```

Top 6 industries (level1&2-excluded, mean):

```
list(industry_group = c("#fitness", "#tech", "#biology", "#legal", "#trade",  
"#energy"))
```

► Code

High Income Industry ranking



Examining the top six industries by both median and mean income (after exclusion), **tech**, **biology**, and **energy** consistently rank near the top and also have sufficiently large sample sizes to draw meaningful conclusions. In contrast, other industries, such as **storage facility**, **chemical**, **tourism**, **fitness**, **legal**, and **trade**, appear in the top ranks but have very small sample sizes. This suggests that either the observed high incomes are not representative of the overall industry, or the industry categorization could be further refined for improved accuracy.

4.2 Age / Work Experience / Education vs Income

In this section, we examine how **age**, **professional work experience**, and **education** influence income levels. We begin by observing the median and mean income across each category:

► Code

age	annual_income_usd_count	annual_income_usd_median
<chr>	<int>	<dbl>
under 18	14	37160
18-24	1223	50000
25-34	12658	70408
35-44	9903	85634
45-54	3188	90000
55-64	992	85000
65 or over	94	87100

7 rows | 1-3 of 4 columns

► Code

education	annual_income_usd_count
<chr>	<int>
High School	639
Some college	2070
College degree	13525
Professional degree (MD, JD, etc.)	1323
Master's degree	8870
PhD	1426
NA	219

7 rows | 1-2 of 4 columns

► Code

years	annual_income_usd_count	annual_income_usd_med
<chr>	<int>	<d
1 year or less	528	5310
2-4 years	3028	5962
5-7 years	4883	6800

years	annual_income_usd_count	annual_income_usd_med
<chr>	<int>	<d
8-10 years	5380	7521
11-20 years	9624	8540
21-30 years	3636	9100
31-40 years	870	9100
41 years or more	123	9400

8 rows | 1-3 of 4 columns

► Code

years_in_field	annual_income_usd_count	annual_income_usd_med
<chr>	<int>	<d
1 year or less	1493	5200
2-4 years	6252	6200
5-7 years	6519	7380
8-10 years	4981	8340
11-20 years	6536	9310
21-30 years	1868	10430
31-40 years	382	10350
41 years or more	41	9800

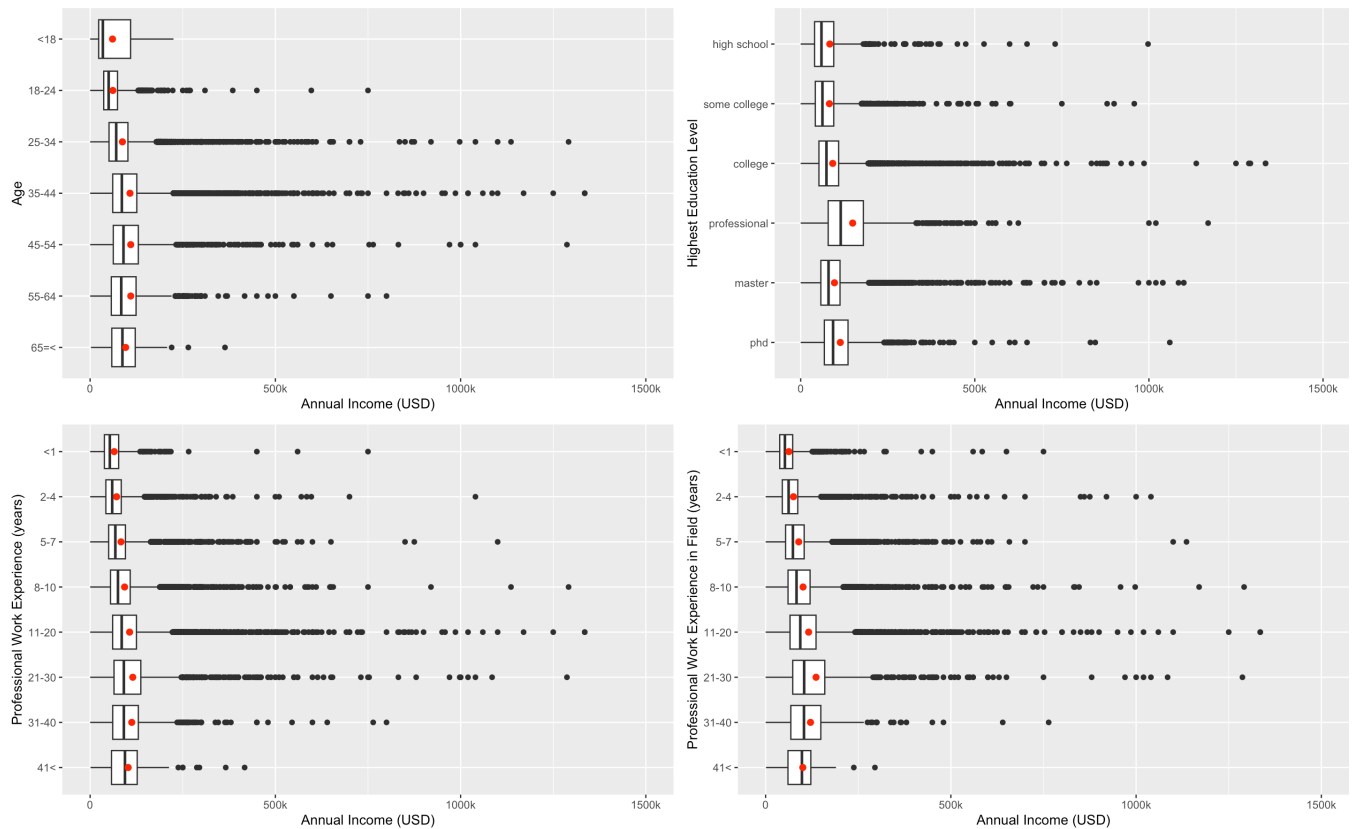
8 rows | 1-3 of 4 columns

Overall, the results appear reasonable, except for the **High School** education category, which shows an unusually high mean income. Upon inspection, there is one record with an annual income exceeding **500,000 USD** whose highest education level is high school. Although this data point may be valid, it lies far beyond the general income range for this education level. Therefore, we will exclude this record from further analysis.

Next, we visualize the **median** and **mean** values using **boxplots** for each category:

► Code

Annual Income Distributions by Age / Education / Work Experience



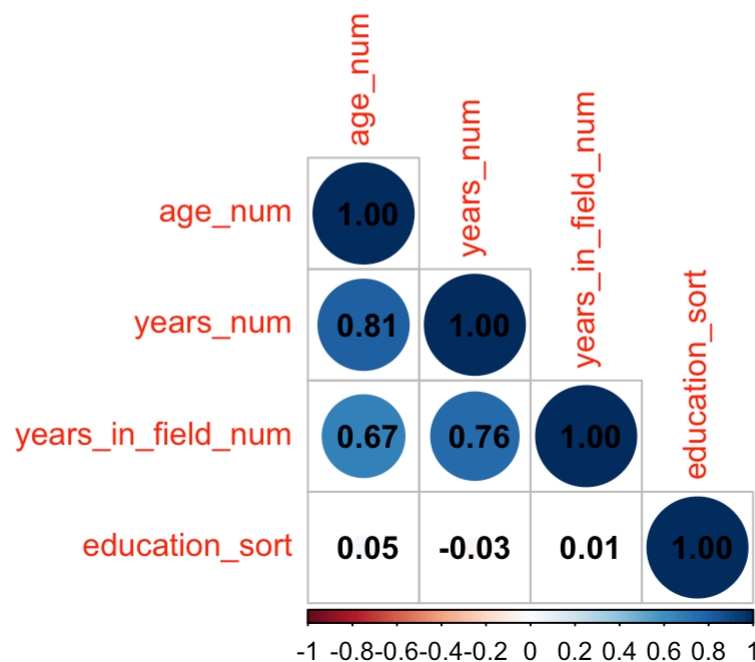
In the plots, red points represent the mean values.

- **Age:** Both the median and mean incomes increase with age, peaking in the **45–54** age group before slightly declining.
- **Education:** Generally, higher educational attainment correlates with higher income. Notably, individuals with **professional degrees** tend to earn more than those with other advanced degrees.
- **Work Experience:** Regardless of field-specific experience, individuals with **21–30 years** of professional experience exhibit the highest income levels.

There appears to be some correlation among these factors. Let's examine the pairwise relationships:

► Code

Correlation Matrix among Age, Work Experience and Education



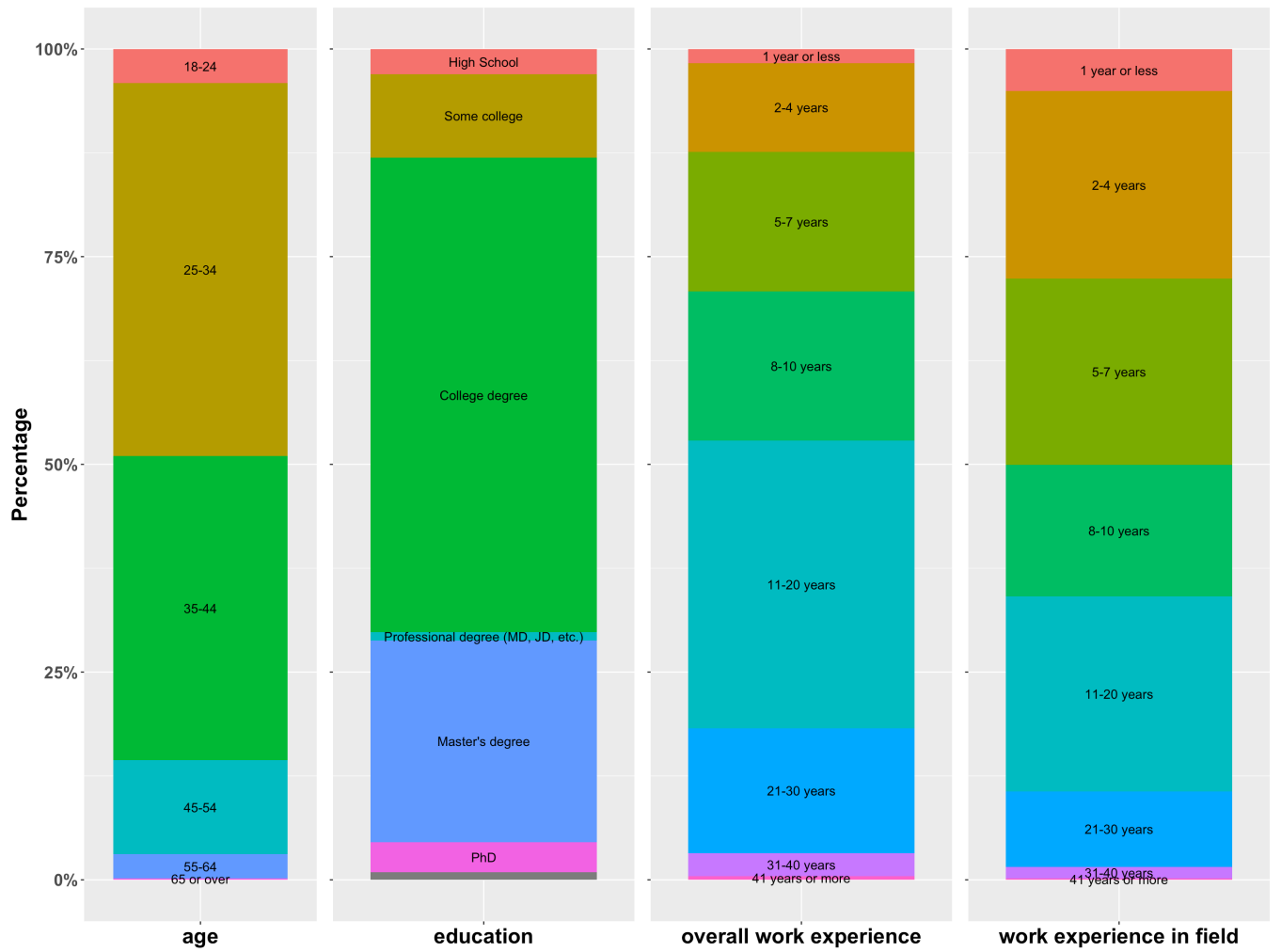
As expected, there are strong positive correlations between **age** and **work experience**, and between the **two types of work experience** themselves. In contrast, correlations between **education** and either **age** or **work experience** are close to zero, suggesting that educational level is relatively independent of these two factors in this dataset.

Referring back to the earlier **Industry vs. Income** analysis, the **tech** industry was identified as one of the highest-paying sectors. To investigate whether this is due to differences in **age**, **work experience**, or **education**, we compare the **tech** industry with **education & training**, a lower-paying industry based on median and mean income.

The proportions of **age**, **work experience**, and **education** categories in both industries are displayed below:

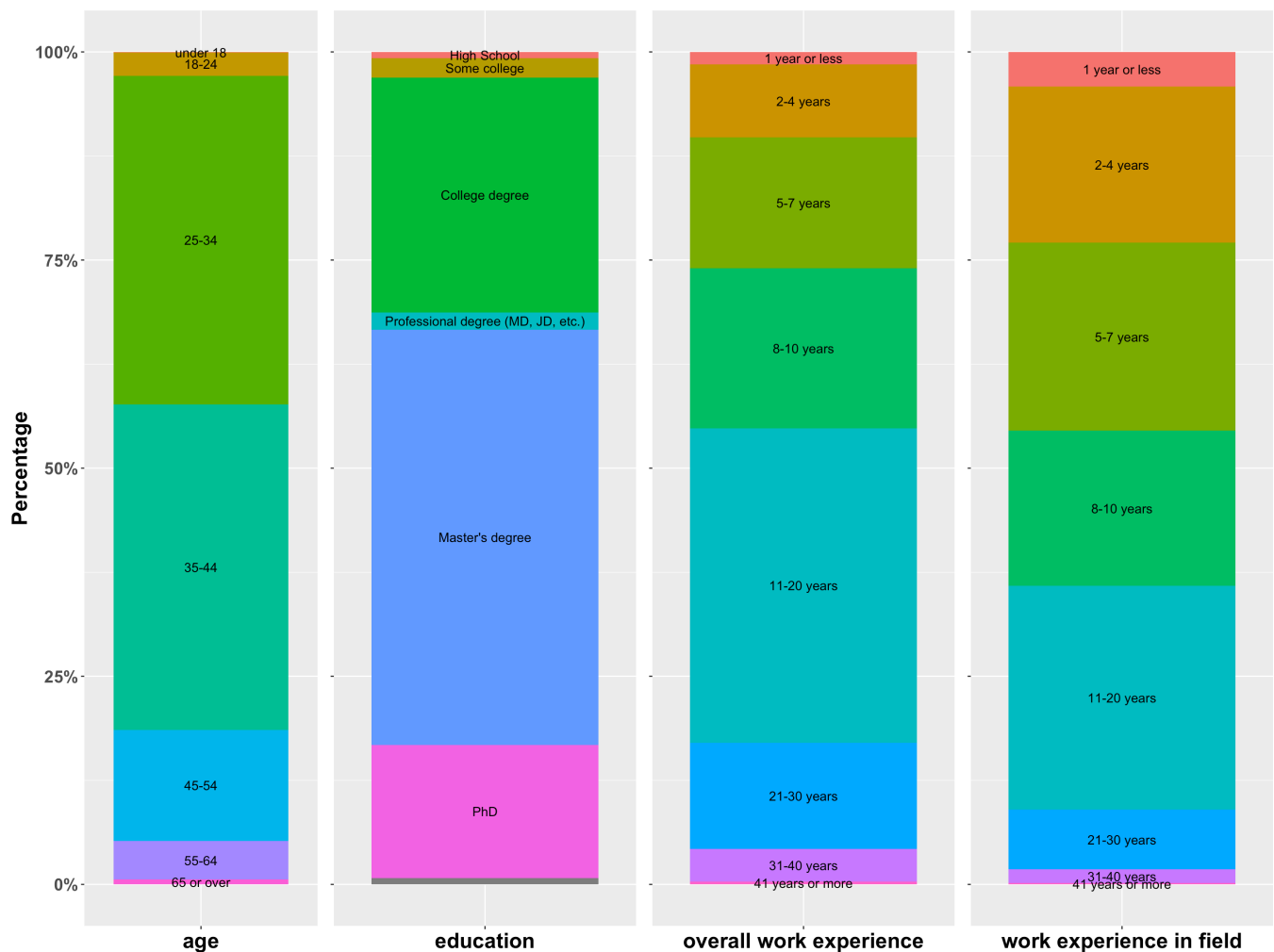
► Code

Proportions of Age / Education / Work Experience in Tech



► Code

Proportions of Age / Education / Work Experience in Education & Training



As shown, the distributions for **age** and **work experience** are quite similar between the two groups. The primary difference lies in education level, Tech professionals are more likely to hold college degrees, whereas Education & Training professionals are more likely to have Master's degrees. Interestingly, individuals with Master's degrees generally earn more than those with college degrees, based on the earlier analysis. Therefore, the income gap between these industries cannot be explained solely by differences in **age**, **experience**, or **education** level. This suggests that industry-specific factors, such as job market demand or business profitability, play a more significant role in income variation.

5 Conclusion

Through this analysis, we explored how **industry**, **age**, **work experience**, and **education** relate to income.

- We identified reasonable patterns, such as higher income with greater experience or higher degrees.
- However, the high-paying industries (e.g., **tech**) do not necessarily employ older, more experienced, or more highly educated individuals.

This indicates that industry characteristics, rather than demographic or educational factors, are likely the main drivers of income differences across professions.

